

LegalEase AI - Core Functionalities & Features

Overview

LegalEase AI provides a complete end-to-end legal drafting lifecycle. Below is a detailed technical examination of each core capability implemented across the frontend and backend.

1. Document Type Selection

- Supported categories include:
 1. Employment Contract
 2. Lease Agreement
 3. Non-Disclosure Agreement (NDA)
 4. Service Agreement
 5. Partnership Agreement
 6. Sales Agreement
 7. Freelance Contract
 8. General Agreement
- Built modularly so additional agreement categories can be introduced simply by extending `SUPPORTED_DOC_TYPES` in `backend/config.py`.

2. Structured Stakeholder Input

- Captures multiple parties, designations, addresses, and corporate status.
- Example:

text

Party 1 (Service Provider): Jane Doe, residing at 123 Tech Avenue, Bengaluru, India

Party 2 (Client): TechNova Solutions Pvt. Ltd., 456 Innovation Park, Mumbai, India

3. Semicolon-Separated Terms & Clauses

- Accepts freeform or bulleted terms separated by semicolons (;).
- Enables intuitive drafting without forcing rigid sub-forms for every clause.
- Example:

```
text
Payment to be made within 30 days of invoice;
Provider agrees to deliver work by the agreed deadline;
Confidentiality must be maintained at all times;
Either party may terminate with 15 days notice
```

4. Input & Date Validation

- Evaluates field lengths, non-emptiness, and semantic consistency.
- Date logic enforces that termination or expiration dates cannot precede commencement or execution dates.
- Raises descriptive errors before triggering expensive AI model calls.

5. AI Legal Drafting

- Synthesizes user specifications into a formal contract using Google Gemini.
- Incorporates preambles, recitals, definitions, substantive clauses, dispute resolution, governing law, and execution blocks.
- Employs bracketed placeholders (e.g. [Jurisdiction]) for omitted jurisdiction-specific items.

6. Styled HTML Document Preview

- Renders the generated legal draft inside an elegant dark-mode container.
- Highlights titles, numbered sections, bold keywords, and bullet points.
- Features prominent legal notices and disclaimers.

7. Interactive Inline Editing

- Toggled via the "**Click to Edit Document**" button.
- Users can directly adjust text, add clauses, or personalize names in a large text area.

- Clicking "**Save Changes**" updates the application state and re-renders previews and download files instantly.

8. Automatic Terms Table Generation

- Parses semicolon-separated terms into clean key-value pairs (e.g. ("Payment", "₹50,000 within 30 days")).
- Renders **Schedule A: Summary of Agreed Key Terms** across:
 - HTML preview table
 - Microsoft Word (.docx) table with slate-navy header shading
 - ReportLab PDF table with alternating row colors and grid borders

9. Plain Text (.TXT) Export

- Formatted as clean UTF-8 text.
- Features standardized ASCII borders, document titles, body clauses, structured terms summary, and the legal disclaimer.

10. Microsoft Word (.DOCX) Export

- Built using `python-docx` .
- Uses formal legal typography (Times New Roman).
- Embeds organization logo centered in the header.
- Includes numbered headings, left-indented sub-clauses, styled Schedule A table, signature blocks, and running footers.

11. Print-Ready PDF (.PDF) Export

- Built with `reportlab` .
- Embeds top branding logo and horizontal accent rules.
- Multi-page canvas automatically stamps running headers, footers, legal disclaimers, and page numbers ("Page X of Y").
- Generates bordered signature lines and date blanks.

12. Custom Branding & Logo Support

- Allows users to upload their organization logo (PNG, JPEG, WebP).
- Automatically scales and embeds the logo across DOCX and PDF exports.
- Gracefully falls back to the default LegalEase logo if no custom file is uploaded.